

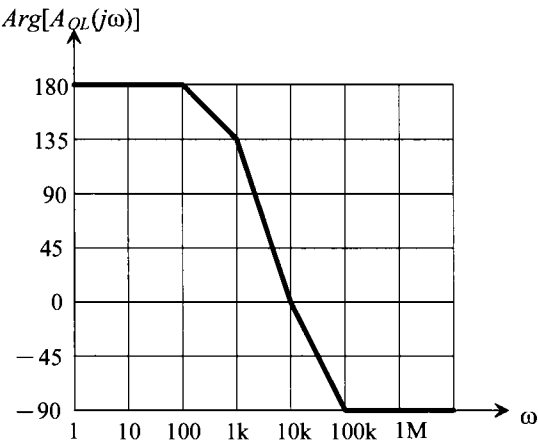


Figure 31.67 Phase response used in problem 31.31.

31.32 You have just measured the gain of the op-amp circuit shown in Fig. 31.68. You know from basic op-amp theory that the gain of the circuit should be $-R_2/R_1$ V/V. However, your measurements with $R_2 = 10\text{ k}\Omega$ and $R_1 = 1\text{ k}\Omega$ revealed that the gain was only -5 V/V . What is the open-loop gain of the op-amp?

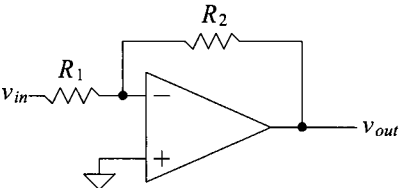


Figure 31.68 How finite open-loop gain effects closed-loop gain, problem 31.32.

31.33 Using the circuit shown in Fig. 31.69 and the *RR* method, find a value of R_1 and A_o which will cause the phase margin to equal 45° at $\omega = 8,000\text{ rad/sec}$. The amplifier can be modeled as having an infinite input impedance and zero output resistance and has a frequency response of

$$A(s) = \frac{-A_o}{(s/200 + 1)(s/10,000 + 1)}$$

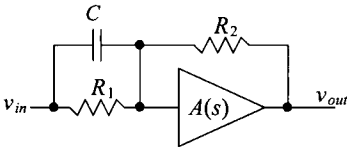


Figure 31.69 Amplifier used in problem 31.33.

- 31.34** Determine if the system with a return ratio as described by Eq. (31.102) is stable.
- 31.35** Redesign the transimpedance amplifier seen in Fig. 31.51 using the long-channel CMOS process discussed in this book.
- 31.36** How would the input-referred noise for the TIA design presented in Sec. 31.9.2 be reduced?

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About the Author

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Square-Law Equations

For a triode-operating long-channel NMOS device

$$I_D = KP_n \cdot \frac{W}{L} \cdot \left[(V_{GS} - V_{THN})V_{DS} - \frac{V_{DS}^2}{2} \right]$$

for $V_{GS} \geq V_{THN}$ and $V_{DS} \leq V_{GS} - V_{THN}$

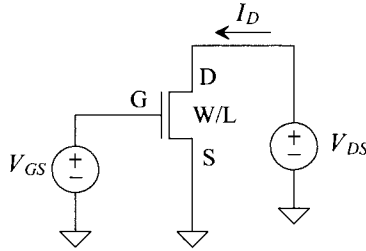
For a long-channel NMOS device operating in the saturation region:

$$I_D = \frac{KP_n}{2} \cdot \frac{W}{L} (V_{GS} - V_{THN})^2 [1 + \lambda(V_{DS} - V_{DS,sat})]$$

for $V_{GS} > V_{THN}$ and $V_{DS} \geq V_{GS} - V_{THN}$

On the border between saturation and triode:

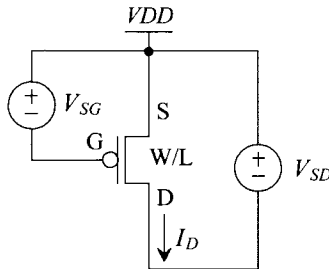
$V_{DS,sat} = V_{GS} - V_{THN}$ and the drain current is called $I_{D,sat}$, see Fig. 6.11



For the PMOS device equations make the following substitutions in the equations listed above

$$V_{DS} \rightarrow V_{SD}, V_{GS} \rightarrow V_{SG}, \text{ and } V_{THN} \rightarrow V_{THP}.$$

All of the voltages and currents in the PMOS and NMOS equations are **positive**. For example, for the PMOS device to conduct a drain current requires $V_{SG} > V_{THP}$. For the NMOS to conduct a drain current requires $V_{GS} > V_{THN}$.



Long-channel MOSFET parameters for general analog design in this book $V_{DD} = 5\text{ V}$ and a scale factor of $1\text{ }\mu\text{m}$ ($scale = 1e-6$)			
Parameter	NMOS	PMOS	Comments
Bias current, I_D	$20\text{ }\mu\text{A}$	$20\text{ }\mu\text{A}$	Approximate
W/L	10/2	30/2	Selected based on I_D and $V_{DS,sat}$
$V_{DS,sat}$ and $V_{SD,sat}$	250 mV	250 mV	For sizes listed
V_{GS} and V_{SG}	1.05 V	1.15 V	No body effect
V_{THN} and V_{THP}	800 mV	900 mV	Typical
$\partial V_{THN,P} / \partial T$	-1 mV/C°	-1.4 mV/C°	Change with temperature
KP_n and KP_p	$120\text{ }\mu\text{A/V}^2$	$40\text{ }\mu\text{A/V}^2$	$t_{ox} = 200\text{ \AA}$
$C'_{ox} = \epsilon_{ox}/t_{ox}$	$1.75\text{ fF}/\mu\text{m}^2$	$1.75\text{ fF}/\mu\text{m}^2$	$C_{ox} = C'_{ox} WL \cdot (scale)^2$
C_{oxn} and C_{oxp}	35 fF	105 fF	PMOS is three times wider
C_{gsn} and C_{sgp}	23.3 fF	70 fF	$C_{gv} = \frac{2}{3}C_{ox}$
C_{gdn} and C_{dgp}	2 fF	6 fF	$C_{gd} = CGDO \cdot W \cdot scale$
g_{mn} and g_{mp}	$150\text{ }\mu\text{A/V}$	$150\text{ }\mu\text{A/V}$	At $I_D = 20\text{ }\mu\text{A}$
r_{on} and r_{op}	$5\text{ M}\Omega$	$4\text{ M}\Omega$	Approximate at $I_D = 20\text{ }\mu\text{A}$
$g_{mn}r_{on}$ and $g_{mp}r_{op}$	750 V/V	600 V/V	Open circuit gain
λ_n and λ_p	0.01 V^{-1}	0.0125 V^{-1}	At $L = 2$
f_{Tn} and f_{Tp}	900 MHz	300 MHz	For $L = 2$, f_T goes up if $L = 1$

Models for digital design using the long- and short-channel processes discussed in this book.				
Technology	R_n	R_p	Scale factor	$C_{ox} = C'_{ox} WL \cdot (scale)^2$
$1\text{ }\mu\text{m}$ (long-channel)	$15k \frac{L}{W}$	$45k \frac{L}{W}$	$1\text{ }\mu\text{m}$	$(1.75\text{ fF}) \cdot WL$
50 nm (short-channel)	$\frac{34k}{W}$	$\frac{68k}{W}$	50 nm	$(62.5\text{ aF}) \cdot WL$

Short-channel MOSFET parameters for general analog design in this book $V_{DD} = 1\text{ V}$ and a scale factor of 50 nm ($scale = 50e-9$)			
Parameter	NMOS	PMOS	Comments
Bias current, I_D	10 μA	10 μA	Approximate, see Fig. 9.31
W/L	50/2	100/2	Selected based on I_D and V_{ov}
Actual W/L	2.5 $\mu\text{m}/100\text{nm}$	5 $\mu\text{m}/100\text{nm}$	L_{min} is 50 nm
$V_{DS,sat}$ and $V_{SD,sat}$ V_{ovn} and V_{ovp}	50 mV 70 mV	50 mV 70 mV	However, see Fig. 9.32 and the associated discussion
V_{GS} and V_{SG}	350 mV	350 mV	No body effect
V_{THN} and V_{THP}	280 mV	280 mV	Typical
$\partial V_{THN,P}/\partial T$	- 0.6 mV/C°	- 0.6 mV/C°	Change with temperature
v_{satn} and v_{satp}	110 x 10 ³ m/s	90 x 10 ³ m/s	From the BSIM4 model
t_{ox}	14 Å	14 Å	Tunnel gate current, 5 A/cm ²
$C'_{ox} = \epsilon_{ox}/t_{ox}$	25 fF/ μm^2	25 fF/ μm^2	$C_{ox} = C'_{ox} WL \cdot (scale)^2$
C_{oxn} and C_{oxp}	6.25 fF	12.5 fF	PMOS is two times wider
C_{gsn} and C_{gsp}	4.17 fF	8.34 fF	$C_{gs} = \frac{2}{3} C_{ox}$
C_{gdn} and C_{gdp}	1.56 fF	3.7 fF	$C_{gd} = CGDO \cdot W \cdot scale$
g_{mn} and g_{mp}	150 $\mu\text{A}/\text{V}$	150 $\mu\text{A}/\text{V}$	At $I_D = 10\text{ }\mu\text{A}$
r_{on} and r_{op}	167 k Ω	333 k Ω	Approximate at $I_D = 10\text{ }\mu\text{A}$
$g_{mn}r_{on}$ and $g_{mp}r_{op}$	25 V/V	50 V/V	!!Open circuit gain!!
λ_n and λ_p	0.6 V ⁻¹	0.3 V ⁻¹	$L = 2$
f_{Tn} and f_{Tp}	6000 MHz	3000 MHz	Approximate at $L = 2$

Effective digital switching resistances and oxide capacitances using the drawn sizes seen for both the long- and short-channel processes.					
Technology	Drawn	Scale factor	Actual size	$R_{n,p}$	$C_{ox,n,p}$
NMOS (long-channel)	10/1	1 μm	10 μm by 1 μm	1.5k	17.5 fF
PMOS (long-channel)	30/1	1 μm	30 μm by 1 μm	1.5k	52.5 fF
NMOS (short-channel)	10/1	50 nm	0.5 μm by 50 nm	3.4k	625 aF
PMOS (short-channel)	20/1	50 nm	1 μm by 50 nm	3.4k	1.25 fF